Software Requirements Specification (SRS)

Student Project Collaboration and Experience Platform

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Preface

This Software Requirements Specification (SRS) document provides a comprehensive description of the **Student Project Collaboration and Experience Platform**, including its purpose, actors, workflows, system rules, scoring mechanisms, governance policies, and operational behavior. The document is structured so that a reader can clearly understand how the system functions and visualize how users interact with the platform from project creation to contribution verification.

The platform is designed to help students collaborate on real-world projects, track work progress, and build verifiable experience records. Rather than hosting development artifacts directly, the platform focuses on project coordination, task tracking, and contribution verification, while the actual implementation work may occur on external development platforms.

The system maintains detailed records of each contribution and uses rule-based automation to award **Experience Points (XP)** and calculate **Reputation Scores**. These mechanisms allow the platform to measure both productivity and reliability of contributors.

The platform also supports mentor participation through recommendation badges and administrative oversight through moderation and audit logging to ensure transparency and trust within the collaboration ecosystem.

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1. Introduction

1.1 Purpose

The purpose of this system is to create a structured environment where students can collaborate on projects while maintaining a transparent record of their work. The platform enables project owners to coordinate teams, assign tasks, and verify contributions. Collaborators are able to participate in projects, perform tasks, and accumulate verifiable experience through approved work.

The system also introduces mechanisms that measure both productivity and reliability. Productivity is measured through Experience Points (XP), while reliability is measured through a Reputation Score. Together these metrics help identify contributors who are both skilled and dependable.

1.2 Scope

The platform supports the following capabilities:

• Project creation and management • Task management through structured workflow states • Contributor collaboration across multiple projects • Verification of external work through evidence links • Automated experience point calculation • Reputation scoring based on contributor behavior • Mentor guidance and recommendation badges • Administrative moderation and governance • Immutable audit logging of system events

The system does not store project source code or development artifacts. Instead, contributors submit links to external evidence of work, such as pull requests or commits, which are then verified by the project owner.

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2. Actors

The system includes four primary actors, each with specific roles and responsibilities.

2.1 Project Owner

A Project Owner is a student who initiates and manages a project. The owner defines the project scope, creates tasks, assigns collaborators, reviews submitted work, and approves or requests revisions.

Responsibilities of a Project Owner include:

• Creating and configuring projects • Defining project objectives and duration • Creating tasks and defining task difficulty levels • Reviewing collaborator applications • Selecting collaborators for the project • Assigning tasks to collaborators • Monitoring project progress • Reviewing submitted work evidence • Approving tasks or requesting revisions • Rating collaborators • Closing projects when work is completed

Project owners verify submitted work externally before approving tasks within the platform.

2.2 Collaborator

Collaborators are students who contribute work to projects. A collaborator may participate in multiple projects simultaneously.

Responsibilities of a collaborator include:

• Exploring available projects • Applying to participate in projects • Accepting assigned tasks • Performing work externally • Submitting evidence of completed work • Responding to revision requests • Monitoring their personal contribution history • Earning experience points for approved tasks

Collaborators interact with the platform primarily to track task assignments and record their completed work.

2.3 Mentor

Mentors are individuals who provide guidance to project teams. Mentors may be faculty members, senior students, or alumni with relevant expertise.

Mentors do not participate in the operational workflow of project approvals. They cannot approve tasks, modify experience points, or influence the scoring system.

Responsibilities of mentors include:

• Providing guidance to project teams • Answering technical questions • Suggesting improvements • Monitoring project progress • Recommending students for future projects • Issuing mentor recommendation badges

Mentor involvement focuses entirely on guidance and recognition rather than evaluation.

2.4 Administrator

Administrators are responsible for ensuring the integrity and proper functioning of the platform.

Responsibilities include:

• Managing user accounts • Investigating suspicious behavior • Reviewing flagged profiles • Issuing warnings to users • Suspending or restricting accounts when necessary • Monitoring system activity • Maintaining audit logs

Administrators ensure that the platform remains trustworthy and that system rules are enforced.

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3. User Roles and Operational Modes

Each account on the platform represents a single user identity. Instead of creating separate accounts for different responsibilities, the platform allows one account to operate in different **modes** depending on the actions performed by the user. These modes correspond to the functional roles within the system.

A single user may therefore act as:

• a collaborator in one project • a project owner in another project • a mentor if approved by the administrator

Roles are therefore **context-based capabilities rather than permanent account types**.

3.1 Default User Mode

When a user first creates an account, they are registered as a standard user. At this stage the user does not possess any elevated privileges such as mentorship authority. However, every user is immediately allowed to either **start their own project** or **join an existing project as a collaborator**.

The default capabilities available to every registered user include:

• creating a new project and becoming the owner of that project • browsing available projects • applying to participate in projects created by other users • joining projects as a collaborator • accepting assigned tasks • submitting evidence of completed work • tracking personal contribution history

Therefore, a newly registered user can naturally act in two capacities:

• **Project Owner** – when they create a project • **Collaborator** – when they join someone else's project

Mentorship privileges are not granted at account creation and require a separate approval process handled by administrators.

3.2 Owner Participation as Collaborator

A project owner may also choose to participate in their own project as a contributor. Ownership and collaboration are treated as separate permissions.

When a project is created the owner is initially registered as:

Owner = true Collaborator = false

If the owner wishes to perform tasks within the project, they may explicitly join the project as a collaborator. Once this occurs the system records:

Owner = true Collaborator = true

The owner can then accept and complete tasks like any other contributor.

3.3 Task Approval Authority

Because the project idea and responsibility belong to the owner, the owner retains approval authority for all tasks within the project, including tasks completed by themselves.

When the owner completes a task and approves it, the system records the approval transparently in the audit log to ensure accountability.

Example audit entry:

Task Approved Submitted By: Project Owner Approved By: Project Owner Approval Type: Self Approval

Experience points are awarded normally for the task. However, the reputation bonus granted for task approval may be reduced to reflect the fact that the approval was self-verified.

3.4 Mentor Mode

Mentorship privileges are not granted automatically during account creation. A user who wishes to act as a mentor must apply for mentor status.

During this process the user submits a professional profile link. Administrators evaluate the applicant using defined eligibility criteria before granting mentor privileges.

Once approved, the user gains access to mentor capabilities including:

• providing guidance to project teams • answering technical questions • suggesting improvements • issuing recommendation badges • recommending capable collaborators to project owners

Mentors operate in an advisory capacity and do not interfere with task approvals or scoring mechanisms.

3.5 Role Interaction Across Projects

Because roles are context-dependent, a user may simultaneously act in multiple modes across different projects.

For example:

User A may be:

• a collaborator in Project X • a project owner in Project Y • a mentor providing guidance to Project Z

This design allows the platform to support natural career progression, where experienced contributors may eventually become mentors while still participating in projects.

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4. Project Lifecycle

Every project on the platform follows a structured lifecycle beginning with project creation and ending with project closure.

3.1 Project Creation

A project begins when a project owner creates a new project and defines the following information:

• Project title • Project description • Project objectives • Estimated duration • Task list • Task difficulty levels

Projects may be configured to allow open applications or to operate through direct invitations.

3.2 Collaboration Process

Students browse available projects and apply to participate. The project owner reviews applications and selects collaborators.

Once accepted into a project, collaborators may be assigned tasks that correspond to the project goals.

3.3 Task Workflow

Each task progresses through a defined set of states.

Task states include:

Open → In Progress → Submitted → Approved / Needs Revision

Open The task exists but has not yet been started.

In Progress A collaborator has accepted the task and is working on it.

Submitted The collaborator has completed the work and submitted evidence.

Approved The project owner has verified the work and accepted the task.

Needs Revision The project owner has reviewed the work and determined that improvements are required.

3.4 Task Execution

Collaborators perform the task work externally using appropriate tools or platforms.

Once the work is completed, the collaborator submits evidence of completion. Evidence may include:

• a pull request link • a commit reference • documentation link • design artifacts

The task status then moves to the Submitted state.

3.5 Task Review

The project owner reviews the submitted evidence and verifies whether the task requirements have been met.

Possible outcomes include:

Approval The task is accepted and XP is awarded.

Revision The collaborator is requested to modify the work and resubmit the task.

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4. Task Tracking Interface

The platform organizes project tasks visually according to their status. Tasks appear within columns representing their current stage in the workflow.

For example:

Open In Progress Submitted Approved

This interface allows both project owners and collaborators to clearly understand the current state of the project and track the progress of individual tasks.

Project owners can quickly see which tasks are pending review and which collaborators are actively contributing.

Collaborators can easily identify the tasks assigned to them and monitor their progress.

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5. Experience Points System

Experience Points (XP) measure the amount of work completed by a collaborator. XP increases when tasks are approved by the project owner.

XP represents productivity and accumulated experience on the platform.

XP values never decrease once earned.

5.1 Base XP Values

Each task is assigned a difficulty level when it is created.

The base XP values are defined as follows:

Easy Task 10 XP

Medium Task 20 XP

Hard Task 40 XP

5.2 Early Completion Bonus

If a collaborator submits a task before the defined deadline, an additional bonus is awarded.

Early completion bonus:

+5 XP

5.3 XP Calculation

XP earned from a task is calculated using the following rule.

XP Earned = Base XP + Early Completion Bonus

The early completion bonus applies only if the task is submitted before the deadline.

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6. Reputation System

The Reputation Score measures the reliability and trustworthiness of collaborators.

Unlike XP, reputation can both increase and decrease depending on contributor behavior.

The reputation system ensures that project owners can identify collaborators who consistently demonstrate responsible behavior.

6.1 Difference Between XP and Reputation

Aspect	Experience Points (XP)	Reputation Score
Purpose	Measures the amount of work completed by a user	Measures the reliability and trustworthiness of a user
How it changes	Increases when tasks are approved	Can increase or decrease based on behavior
Main trigger	Task completion and early submission	Approval outcomes, revisions, deadlines, and mentor endorsements
Decrease possible	No, XP does not decrease once earned	Yes, reputation may decrease due to negative behavior
Platform use	Used for leaderboard ranking and experience tracking	Used to evaluate collaborator reliability and professional conduct

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6.1 Positive Reputation Rules

Task Approved

When a collaborator successfully completes a task and the task is approved by the project owner:

+8 reputation

Mentor Recommendation Badge

When a mentor issues a recommendation badge to a collaborator:

+20 reputation

6.2 Negative Reputation Rules

Task Revision Required

When a task submission requires revision:

-3 reputation

Deadline Missed

If a task is submitted after its deadline:

-5 reputation

Leaving Project Midway

If a collaborator abandons a project after accepting tasks:

-20 reputation

Exception: If the project owner approves the withdrawal, the penalty does not apply.

Administrator Warning

If an administrator issues a warning due to misuse or misconduct:

-15 reputation

6.3 Reputation Decay

To ensure that reputation reflects current activity levels, the system gradually reduces reputation for inactive users.

If a user remains inactive for 30 days:

Reputation = Reputation × 0.98

Reputation decay stops immediately when the user becomes active again.

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7. Mentor Recommendation Badges

Mentors may endorse collaborators through recommendation badges. These badges serve as recognition of positive contributions and professional conduct.

Examples of badges include:

• Recommended Contributor • Reliable Team Member • Strong Problem Solver

Recommendation badges increase reputation but do not influence experience points.

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8. Contribution Ledger

Every approved task generates a permanent contribution record. This record forms part of the collaborator's verified work history.

Each contribution record includes the following information:

• User ID • Project ID • Task ID • Evidence link • Approval timestamp • XP awarded

The contribution ledger ensures transparency and allows collaborators to demonstrate verified project experience.

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9. User Dashboard and Project Health Dashboard

Every user on the platform has access to a **personal dashboard** that provides a summary of their activity, contributions, and performance metrics on the platform.

The user dashboard includes a **Contribution Report** which summarizes the work completed by the user across all projects. This report allows users to clearly see their verified contributions and track their progress over time.

The contribution report focuses specifically on **past project work and verified contributions** rather than platform metrics. It acts as a detailed history of the projects a user has participated in and the specific tasks they completed within those projects.

The contribution report includes:

• list of past projects the user has worked on • tasks completed within each project • brief description of the work performed • evidence links for the work (such as pull request or commit links) • approval records for each task

This allows other users and project owners to clearly understand the type of work the collaborator has performed in previous projects.

Experience Points (XP) and Reputation Score are displayed separately on the user dashboard as **visual summary metrics**, such as counters or graphical indicators, allowing users to quickly see their overall performance while the contribution report provides detailed historical context of their work.

Project Health Dashboard

The system provides analytics that help project owners and mentors monitor project progress.

Metrics may include:

• Number of completed tasks • Number of pending submissions • Active collaborators • Percentage of project completion

These metrics help identify project bottlenecks and evaluate overall progress.

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10. Audit Logging

The platform maintains an immutable audit log that records significant system events.

Events recorded include:

• Task approvals • Reputation updates • XP awards • user warnings • account suspensions

Audit logs cannot be modified after they are recorded.

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11. Functional Requirements

The following functional requirements define the core capabilities that the platform must provide.

FR1 – User Registration The system shall allow users to create accounts by providing basic profile information and a LinkedIn profile link.

FR2 – Project Creation The system shall allow any registered user to create a project and become the project owner for that project.

FR3 – Project Participation The system shall allow users to apply to projects created by other users and participate as collaborators once approved.

FR4 – Task Management The system shall allow project owners to create tasks, define difficulty levels, assign collaborators, and manage task status transitions.

FR5 – Task Submission The system shall allow collaborators to submit evidence of completed work through external links.

FR6 – Task Approval The system shall allow project owners to review submissions and either approve the task or request revisions.

FR7 – Experience Point Allocation The system shall automatically calculate and award experience points when a task is approved.

FR8 – Reputation Updates The system shall update reputation scores according to the defined reputation rules after task approval or negative events.

FR9 – Contribution Reporting The system shall maintain a record of verified contributions and display them in the user dashboard.

FR10 – Leaderboard Generation The system shall generate leaderboards based on XP earned by users.

FR11 – Mentor Recommendation Badges The system shall allow approved mentors to issue recommendation badges to collaborators.

FR12 – Administrative Governance The system shall allow administrators to review mentor applications, monitor platform activity, and issue warnings if necessary.

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12. Assumptions and Constraints

Assumptions

The system operates under the following assumptions:

• users have access to external development platforms where project work is performed • collaborators submit valid evidence links when reporting completed work • users provide authentic LinkedIn profiles during registration • project owners responsibly verify submissions before approval

Constraints

The system operates with the following constraints:

• development artifacts such as source code are not stored directly within the platform • work verification relies on external evidence links • scoring mechanisms must remain deterministic and rule-based • users cannot generate XP without an approved task

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13. Non Functional Requirements

NFR1 –Usability

The system must provide an interface that allows users to easily understand project status and task assignments.

NFR2 –Security

The platform must enforce role-based access control to ensure that users can perform only the actions allowed by their role.

NFR3 –Reliability

XP transactions must execute exactly once to prevent duplicate scoring.

NFR4 –Scalability

The system must support multiple projects and concurrent users without performance degradation.

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12. XP Leaderboard

The platform includes a leaderboard that ranks collaborators based on the total XP they have earned through approved work. The leaderboard provides visibility into the most active contributors on the platform and encourages healthy competition among students.

The leaderboard is recalculated periodically using the total accumulated XP of each collaborator.

Information displayed on the leaderboard may include:

• Username • Total XP • Reputation score • Number of completed tasks • Number of projects participated in

The leaderboard helps highlight highly productive contributors and motivates users to remain active on the platform.

Leaderboard rankings are determined strictly by XP and are not directly affected by reputation scores. Reputation remains a trust indicator, while XP reflects productivity.

To ensure fairness, only approved tasks contribute to XP calculations. Tasks that are rejected or still under review do not affect leaderboard rankings.

Seasonal Leaderboards

To ensure fairness and maintain engagement for new users, the system may also provide **seasonal leaderboards** based on shorter time periods.

Examples include:

• Weekly leaderboard • Monthly leaderboard • All-time leaderboard

Seasonal leaderboards highlight recent activity, allowing new contributors to compete with experienced users and preventing rankings from being permanently dominated by early adopters.

XP Fairness Rules

To prevent score inflation and misuse of the scoring system, the platform follows these rules:

• XP is awarded **only once per task after approval**. • Duplicate submissions cannot generate additional XP. • XP values are fixed according to task difficulty and cannot be arbitrarily increased. • Tasks that are rejected or still under review do not contribute to leaderboard rankings.

These rules ensure that leaderboard rankings reflect genuine work contributions rather than repeated or manipulated submissions.

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13. Mentor Eligibility and Approval Rules

Mentors play an advisory role in the platform by guiding student teams, answering questions, and recommending capable collaborators. Because mentors influence learning and project direction, the platform requires a verification process before granting mentor privileges.

Unlike collaborators, mentors may not necessarily be active project contributors on the platform. Therefore mentor approval is based primarily on **professional experience and external credentials** rather than platform reputation.

During account creation, users who wish to become mentors must provide a **LinkedIn profile link**. Administrators review this profile to determine whether the applicant meets the eligibility requirements for mentorship.

The administrator evaluates the candidate using the following criteria.

13.1 Professional Experience

The candidate should have a minimum of **4 to 5 years of professional experience** in a relevant technical or professional domain such as software development, system design, data science, product development, or related engineering fields.

This ensures that mentors have sufficient real-world exposure to guide student projects effectively.

13.2 Verified Professional Profile

The LinkedIn profile must appear authentic and complete. Indicators considered during verification include:

• clearly listed work experience • consistent career history • endorsements or recommendations • connections within professional networks

Profiles that appear incomplete or suspicious may be rejected.

13.3 Professional Conduct

The administrator must verify that the applicant demonstrates professional conduct in their public profile and career history. Candidates with controversial or inappropriate professional records may be rejected.

Mentor Privileges

Once approved, mentors gain the ability to:

• provide guidance to project teams • answer technical questions • suggest improvements to project design • issue mentor recommendation badges to collaborators • recommend reliable collaborators to project owners

Mentors **do not** approve tasks, assign XP, or modify reputation scores. Their role is advisory rather than evaluative.

Mentor Revocation

Administrators may revoke mentor privileges if:

• the mentor misuses their authority • the mentor provides misleading guidance • the mentor violates platform policies

Revocation ensures that the mentorship system remains trustworthy and beneficial to student collaborators.

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14. Future Enhancements

Potential future improvements include:

• automated integration with development platforms • advanced analytics dashboards • institutional project tracking • recommendation systems for collaborator selection • implementation of an automated reputation decay mechanism

Reputation Decay Enhancement

In future versions of the platform, an automated **reputation decay system** may be introduced to ensure that reputation reflects **recent activity and current reliability** rather than historical contributions alone.

Under this mechanism, if a user remains inactive on the platform for a defined period (for example, 30 days without task activity or project participation), the system would gradually reduce their reputation score by a small percentage.

Example concept:

Reputation = Reputation × 0.98 after each inactivity period

This feature ensures that active contributors maintain higher trust visibility while users who become inactive slowly lose reputation influence until they resume participation.

The decay process would immediately stop once the user becomes active again, ensuring that reputation accurately reflects ongoing engagement on the platform.

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15. Conclusion

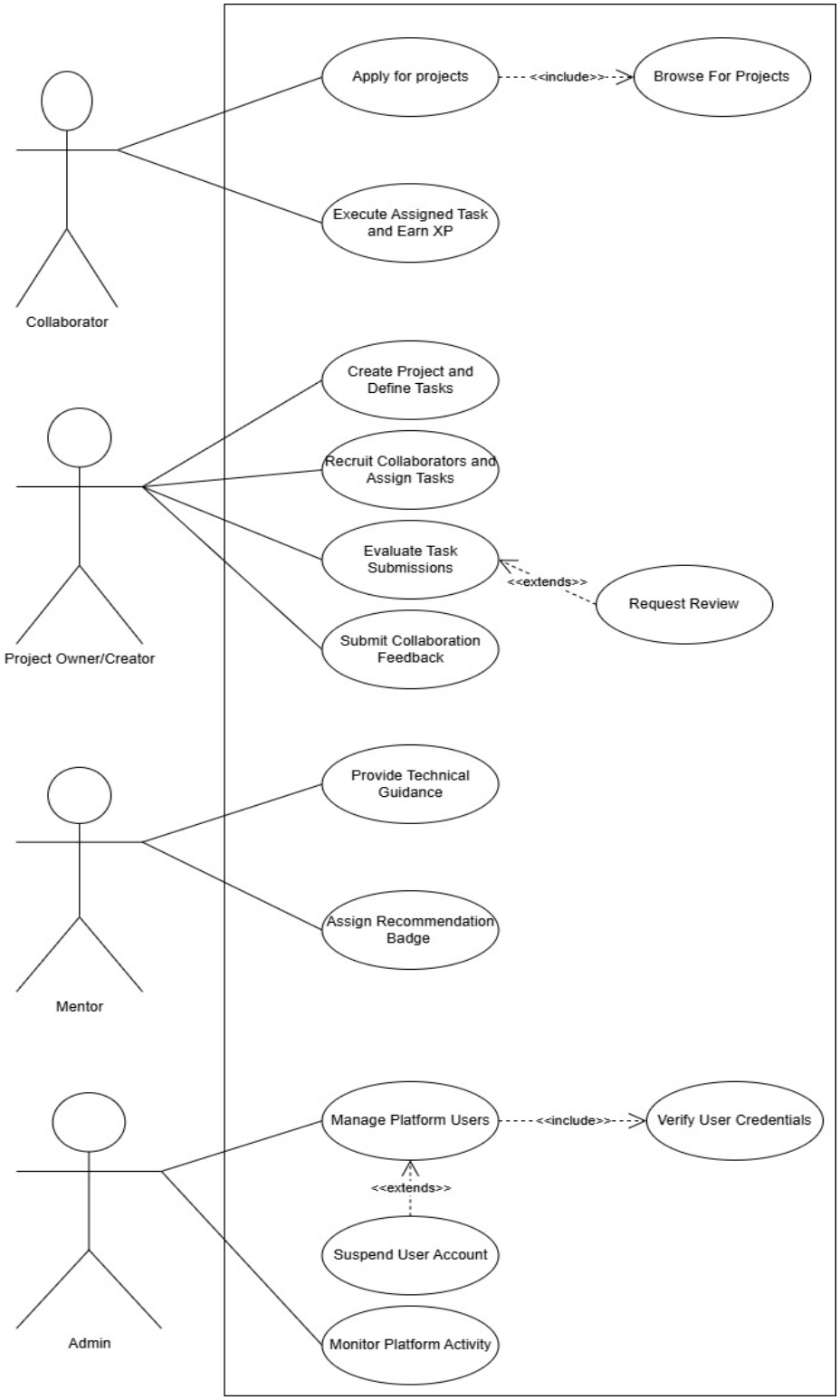
The Student Project Collaboration and Experience Platform provides a structured environment where students can collaborate on projects while building verifiable experience records. The system enables users to create projects, participate as collaborators, receive mentorship guidance, and track their contributions through transparent task workflows.

Through the integration of experience points (XP), reputation scoring, mentor recommendation badges, contribution reports, and a public leaderboard, the platform encourages productive collaboration while maintaining accountability and fairness. The contribution ledger and audit logging mechanisms ensure that work approvals and scoring updates remain transparent and traceable.

By allowing users to operate in multiple roles across different projects—such as project owner, collaborator, and mentor—the platform supports natural growth and progression within the community. Students can begin by contributing to projects, later lead their own initiatives, and eventually qualify to mentor others.

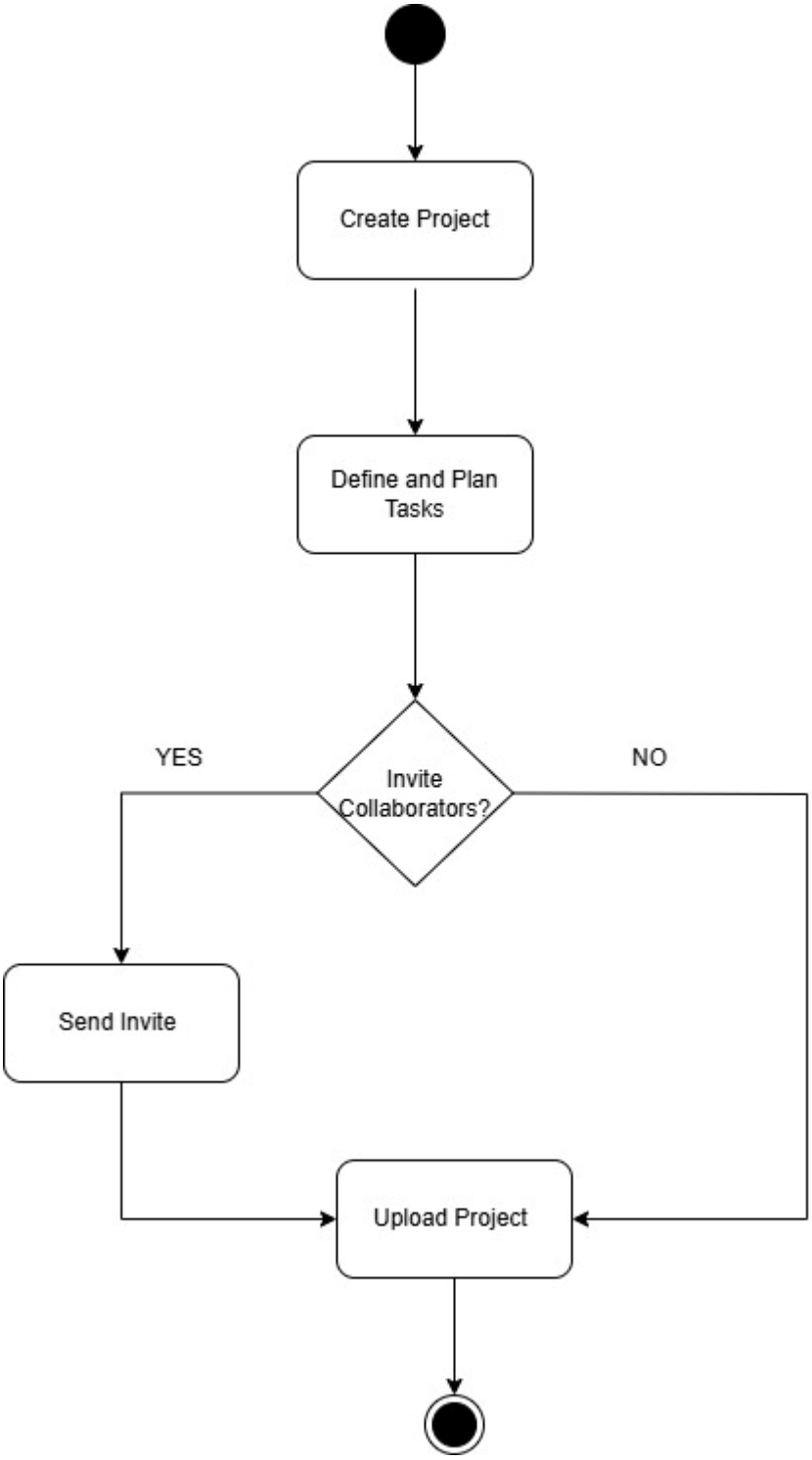
Overall, the system promotes experiential learning, responsible teamwork, and credible documentation of project work. The platform therefore serves not only as a collaboration tool but also as a reliable record of real-world experience that supports academic development and professional growth.

USE CASE DIAGRAM-

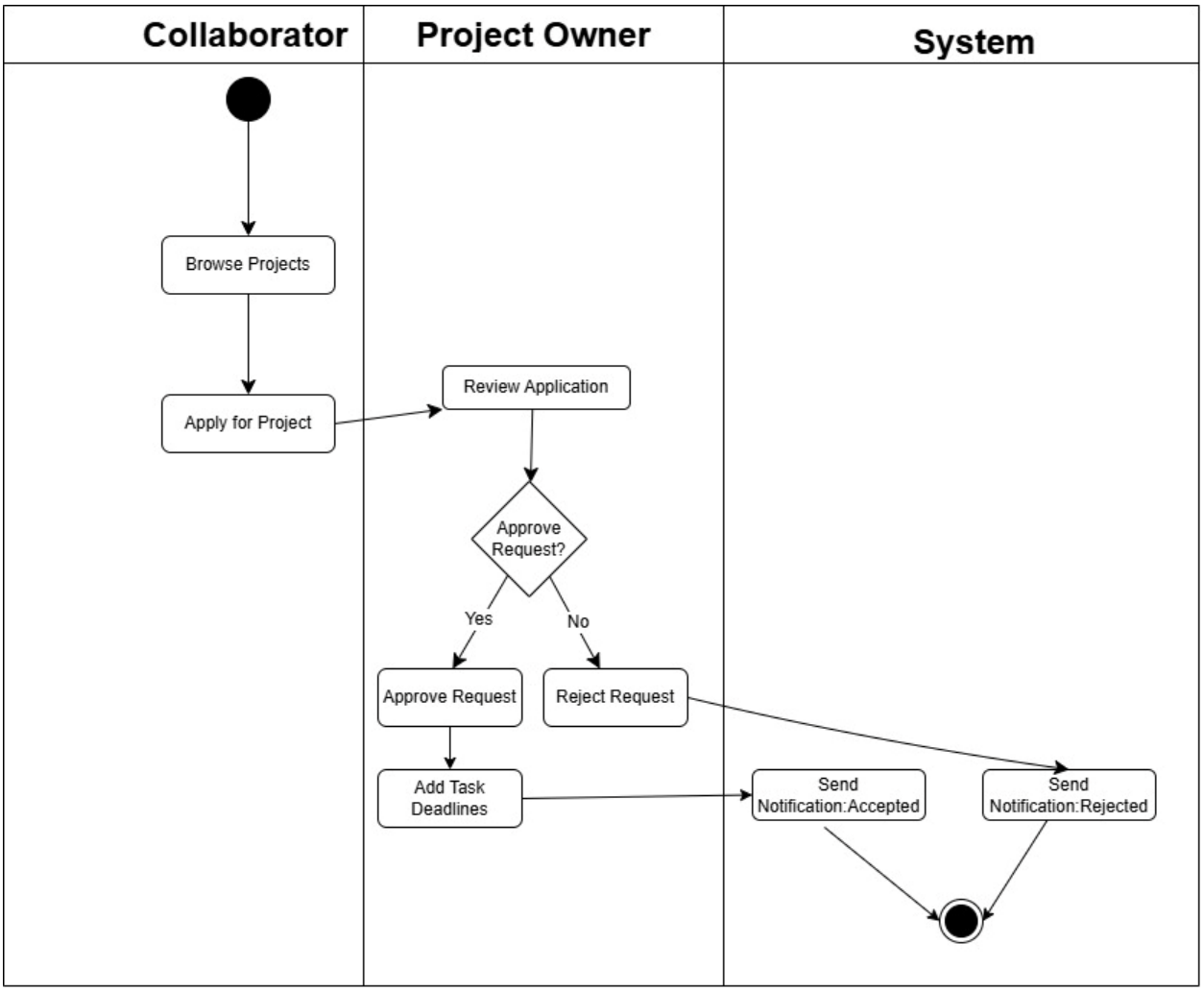


ACTIVITY DIAGRAMS

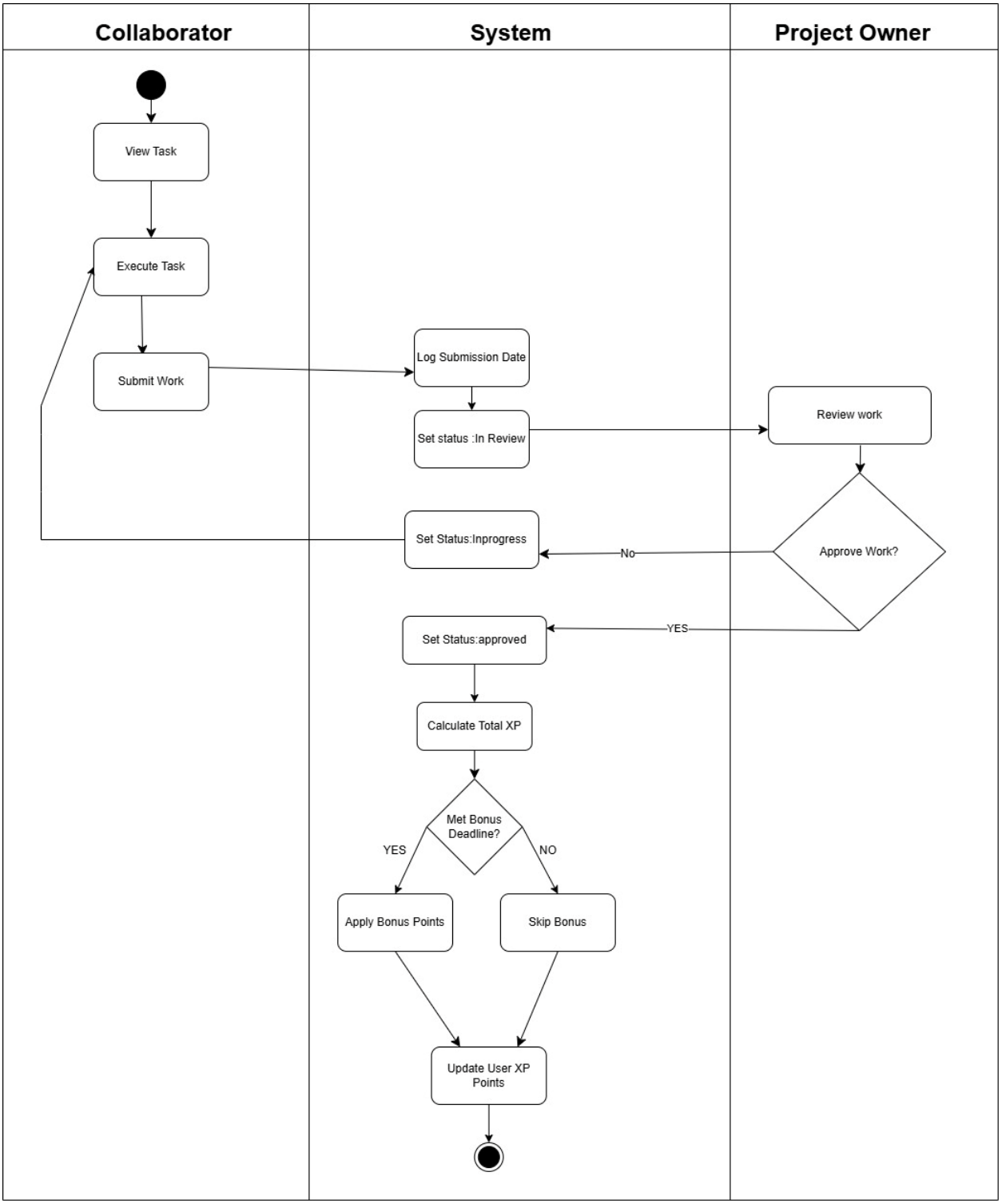
PROJECT CREATION



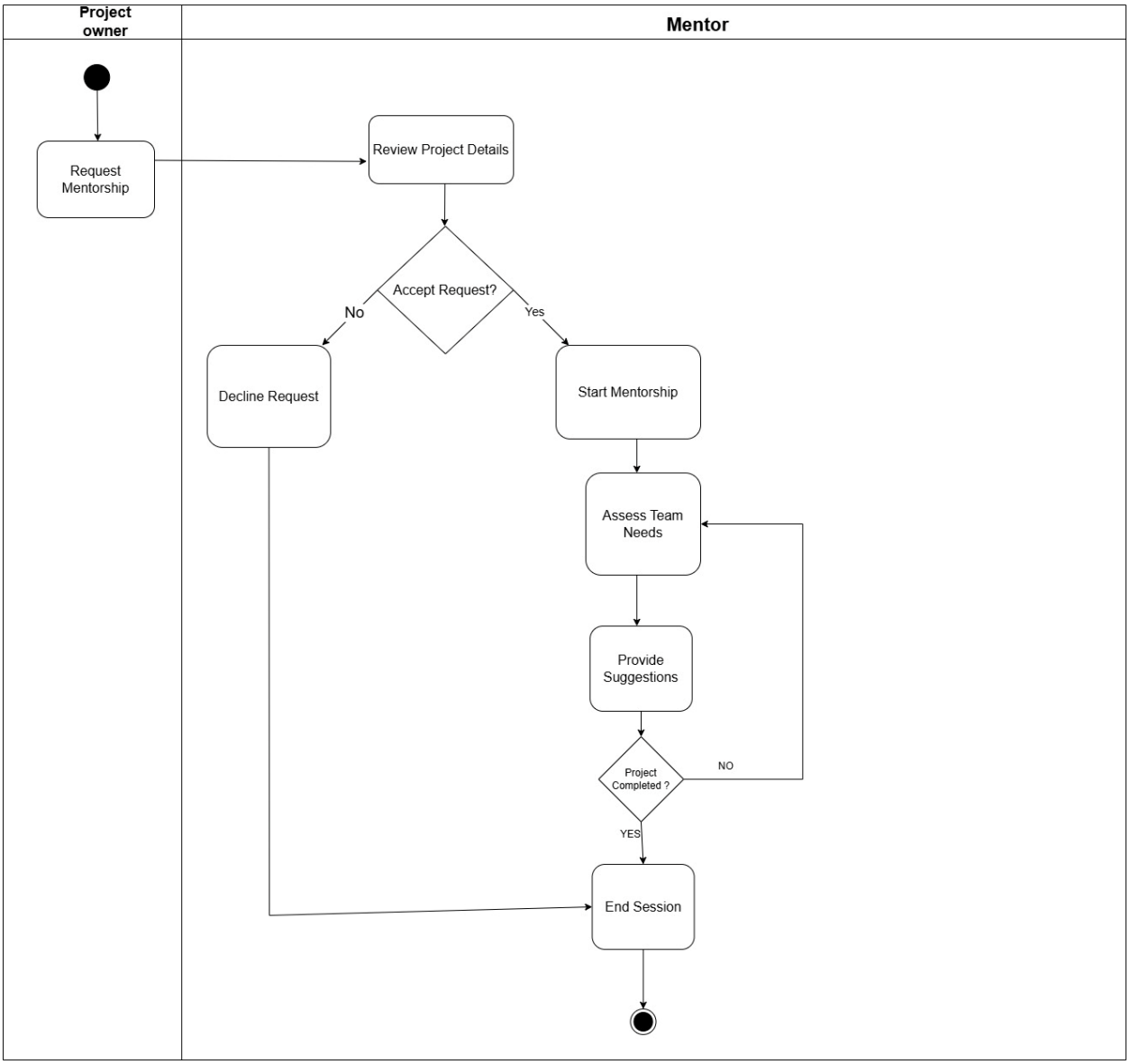
RECRUITMENT



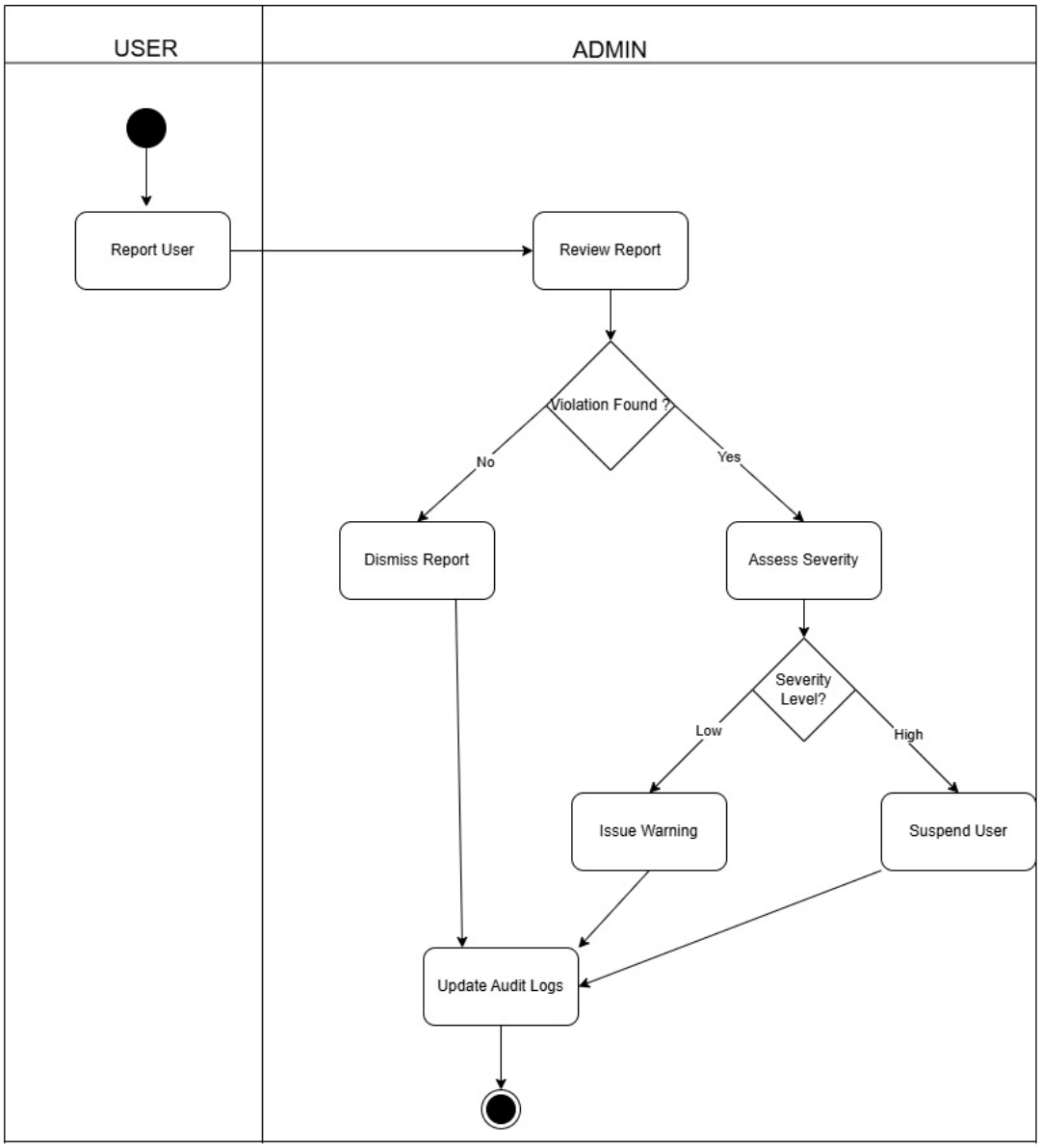
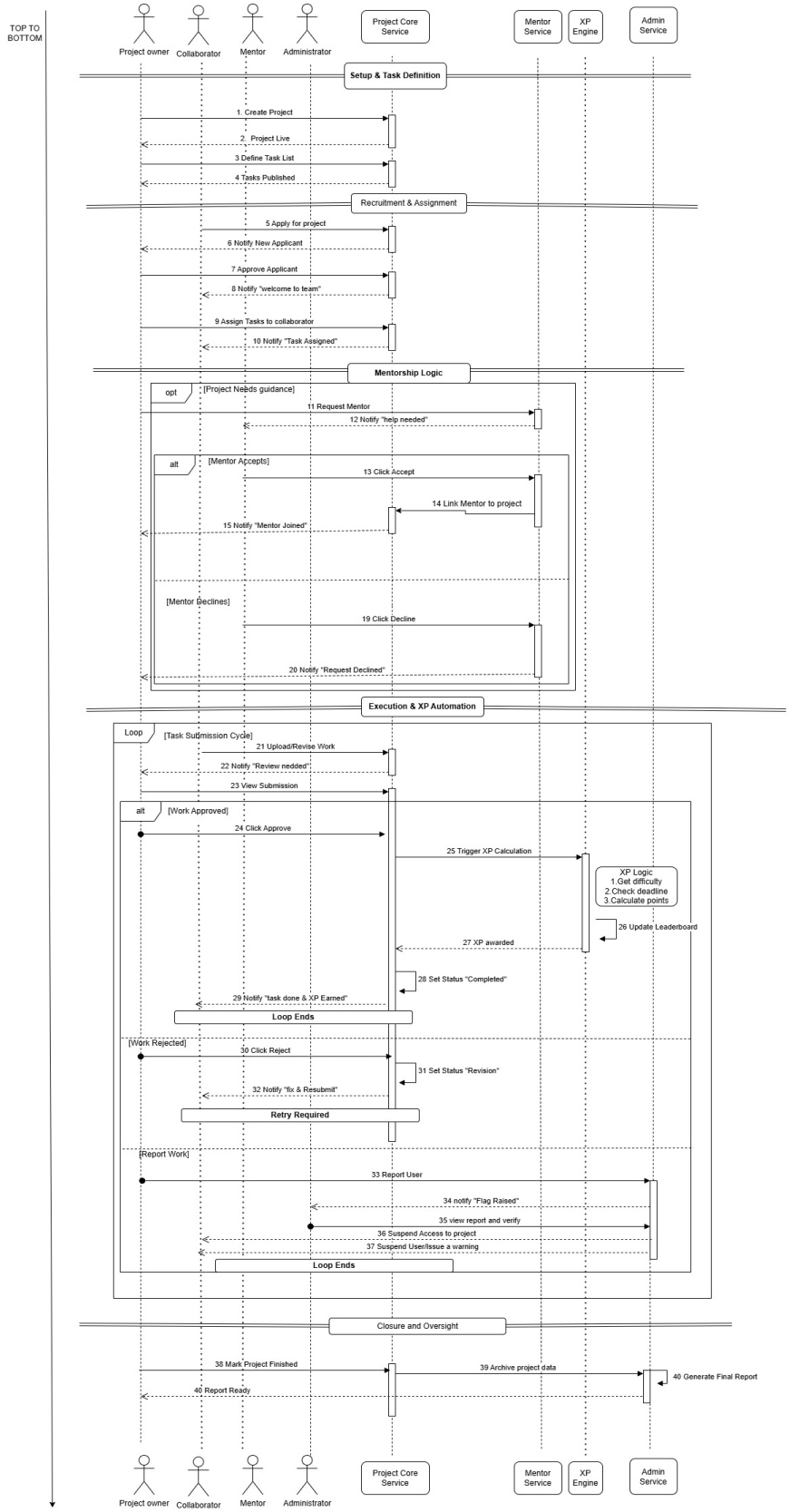
TASK EXECUTION



MENTORSHIP WORKFLOW



ADMIN WORKFLOW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